Quiz #2
EGRE246

Name: _____

1. Write a C program to create a dynamic array to store a list of integer numbers and print out the sum of all the integers. Your program will accept **keyboard inputs** for the total number of integers and a list of integers. It is also expected to print out the integers and the **sum** of all the integers in the end (4 points).

```
#include <stdio.h>
#include <stdlib.h>
void main(){
    int *arr; int i;
    int n, total=0;
    printf("Total number of integer");
    scanf("%d", &n);
    arr=(int *) malloc(n *sizeof(i));
    for(i=0; i<n; i++){scanf("%d", &arr[i]);
                       total += arr[i];}
    for (i=0; i<n; i++) printf("%5d", arr[i]);
        printf("%d\n", total);
}
```

2. Write a C function to swap two variable values (3 points).

```
void swap (int *a, int *b) {
    int temp;
    temp = *a;
    *a = *b;
    *b = temp;
}
```

3. Write a C function to calculate the cubic root of a number with Newton-Raphson method. Assume the number is a positive number, you do not need to deal with the negative number (3 points).

```
double cubicRoot(long x) {
    double error=0.0001;
    int i;
    double s=1, temp=0;
    for (i<0; i< 100,000; i++) {
        temp =s;
        s = (2*s + x/(s*s))/3;
        printf("%5f\t", s);
        if (fabs(s-temp)<=error) break;
    }
    return s;
}
```